

Mohan Krishna Thiriveedhi

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EDUCATION

Kennesaw State University

Master of Science in Computer Science

RVR & JC College Of Engineering

Bachelor of Technology in Computer Science

Aug 2024 - May 2026

CGPA: 4.0/4.0

Aug 2019 - June 2023

CGPA: 9.21/10

SKILLS

Python , Java , C++ , SQL ,Data Structures, Power BI , HTML ,CSS ,Image Processing , Machine Learning, Selenium, Hadoop , Spark, Tensorflow, PyTorch, HTML, CSS, Spark, Flask.

EXPERIENCE

Kennesaw State University | *Graduate Teaching Assistant*

Jan 2025 – Present

- Teaching Python, Java and C Sharp to undergraduate students through In-person labs.
- Grade lab assignments and projects while offering constructive feedback to support learning progress.
- Support over 100 students per semester in building strong programming foundations and problem-solving skills.

Tata Consultancy Services | *Software Engineer Full Time*

June 2023 – July 2024

- Developed 10+ interactive Power BI reports to monitor key performance indicators (KPIs) for Tata Steel Project, improving decision-making processes.
- Optimized SQL data models and Power BI dashboards, cutting refresh times by 30%.
- Ensured data accuracy and integrity in Power BI reports through thorough testing and validation processes.

Peoplehum Technologies Private Limited | *SDET Internship*

Jan 2023 – June 2023

- Automated over 150 test scenarios for iOS and web applications using Selenium and Java.
- Increased iOS automation test coverage from 30% to 95%, validating critical user workflows.
- Performed API load testing with JMeter to ensure scalable data transfer across REST services.

PROJECTS

AI-assisted Diabetic Retinopathy Screening from Fundus Images - [GitHub Repository](#)

- *Python, Pytorch, Pycharm*

- Developed deep learning system achieving 99-100% accuracy for severe diabetic retinopathy classification using ResNet-50, EfficientNet-B0, and ViT on 976-image APTOS dataset with PyTorch and Grad-CAM
- Engineered automated screening pipeline with CLAHE preprocessing and augmentation, achieving perfect AUC (1.0) and F1-scores across 3 architectures in 6 epochs using AdamW optimizer
- Implemented Grad-CAM explainability across 3 deep learning architectures to validate pathological region learning and detect shortcut behavior in Vision Transformer models.
- Developed Flask web application with real-time image upload and prediction interface, displaying DR classifications with Grad-CAM heatmap overlays for clinical interpretation

Large-Scale Cybersecurity Threat Detection - [GitHub Repository](#)

- *Python, PySpark, VS Code*

- Designed a scalable intrusion detection system using PySpark and machine learning models on the UNSW-NB15 dataset, achieving approximately 98% accuracy.
- Applied Principal Component Analysis to reduce features by 49%, resulting in a 40% improvement in training speed without accuracy loss.
- Processed over 2.5 million network traffic records using distributed computing, enabling real-time cybersecurity threat detection.
- Integrated Quantum Machine Learning to improve classification accuracy for complex cyberattack scenarios.

ACHIEVEMENTS

- Secured 2998 Global Rank in TCS Codevita Season 10
- Top SQL 50 Badge in Leetcode